

torch.nn.utils.parametrizations.weight_norm

https://pytorch.org/docs/stable/generated/torch.nn.utils.parametrizations.weight_norm.html

Description:

Apply weight normalization to a parameter in the given module. Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name with two parameters: one specifying the magnitude and one specifying the direction. By default, with `dim=0`, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use `dim=None`. See <https://arxiv.org/abs/1602.07868> module (Module) – containing module

Function Signature

```
torch.nn.utils.parametrizations.weight_norm(module,  
name='weight', dim=0)[source]#
```

Parameters

Returns

Returns:

The original module with the weight norm hook

Code Examples

```
>>> m = weight_norm(nn.Linear(20, 40), name='weight')
>>> m
ParametrizedLinear(
  in_features=20, out_features=40, bias=True
  (parametrizations): ModuleDict(
    (weight): ParametrizationList(
      (0): _WeightNorm()
    )
  )
)
>>> m.parametrizations.weight.original0.size()
torch.Size([40, 1])
>>> m.parametrizations.weight.original1.size()
torch.Size([40, 20])
```

Detailed Documentation

Documentation

Apply weight normalization to a parameter in the given module.

Weight normalization is a reparameterization that decouples the magnitude of a weight tensor from its direction. This replaces the parameter specified by name with two parameters: one specifying the magnitude and one specifying the direction.

By default, with `dim=0`, the norm is computed independently per output channel/plane. To compute a norm over the entire weight tensor, use `dim=None`.

See <https://arxiv.org/abs/1602.07868>

`module (Module)` – containing module

`name (str, optional)` – name of weight parameter

`dim (int, optional)` – dimension over which to compute the norm

The original module with the weight norm hook